

BUG TRACKING SYSTEM

ABSTRACT

The Bug Tracking System is a desktop-based application developed using Java Swing and MySQL as the backend database. It aims to streamline the process of reporting, assigning, tracking, and resolving software bugs during the development lifecycle. The system allows multiple roles such as Admin, Project Manager, Tester, and Developer to interact with the system according to their permissions.

OBJECTIVES

The main objectives of this Bug Tracking System are:

1. To automate the bug tracking process replacing manual tracking methods with a digital, database-driven solution.
2. To provide role-based access control for Admins, Project Managers, Testers, and Developers.
3. To facilitate easy reporting and management of software bugs during the development process.
4. To improve communication between testers and developers through clear bug assignment and status updates.
5. To maintain a centralized database for users, bug reports, and project details.
6. To ensure data consistency and integrity via SQL database connectivity.
7. To enable real-time data viewing using GUI tables populated from SQL queries.
8. To allow for easy maintenance and extension, ensuring scalability for larger teams or projects.

SYSTEM REQUIREMENTS

- Operating System: Windows 10 or 11
- Programming Language: Java
- IDE: NetBeans / Eclipse / IntelliJ IDEA
- Database: MySQL Server 8.0
- Connector: MySQL JDBC Connector (mysql-connector-java.jar)

- Libraries Used: Java Swing (javax.swing), AWT (java.awt), JDBC (java.sql)

MODULE DESCRIPTION

The system is composed of several interlinked modules, each performing a specific role.

- The Admin manages users and roles
- The Tester reports and verifies bugs
- The Project Manager assigns bugs to developers
- The Developer updates bug statuses once fixed.

This project provides an efficient way to centralize bug management and ensure accountability throughout the software development process.

Login Module

- **Class:** Login.java
- **Purpose:** Allows users to log in using their credentials and role.
- **Functions:**
 - Verifies user credentials from the database.
 - Redirects the user to the appropriate window (Admin, Tester, Developer, Project Manager).
 - Includes a “Back” button to return to the main menu.
- **Database Tables Used:** users

Admin Module

- **Class:** AdminWindow.java
- **Purpose:** Central control for managing users and viewing developer/tester information.
- **Functions:**
 - Add, view, and manage users.
 - View all developers and testers.
 - Launch other windows like “View Developers” or “View Testers”.
- **Database Tables Used:** users

Tester Module

- **Class:** TesterWindow.java, AddBug.java, ViewBugs.java
- **Purpose:** Enables testers to log bugs and verify them after resolution.
- **Functions:**
 - Add new bug reports.
 - View all bugs logged by testers.
 - Search and filter bugs by name or status.

- **Database Tables Used:** bugs, users

Project Manager Module

- **Class:** AssignBug.java
- **Purpose:** Assigns bugs to specific developers.
- **Functions:**
 - View list of unassigned bugs.
 - Assign bugs to developers via combo boxes populated from the database.
 - Update bug status in the database.
- **Database Tables Used:** bugs, users

Developer Module

- **Class:** DevelopersWindow.java
- **Purpose:** Displays bugs assigned to the developer and allows them to mark as fixed.
- **Functions:**
 - Retrieve assigned bugs using SQL queries.
 - Update bug status to “Resolved” or “In Progress”.
 - Option to navigate back to main menu.
- **Database Tables Used:** bugs

Database Connectivity Module

- **Class:** DBConnection.java
 - **Purpose:** Establishes and manages connections between the Java application and MySQL.
 - **Functions:**
 - Connects to the MySQL server using JDBC.
 - Provides reusable methods for executing queries.
 - **Key API Used:** DriverManager.getConnection()
-

DATABASE DESIGN

The database 'bugtrackingsystem' stores user details, stores bugs, assign bugs.

Create TABLE users:

```
CREATE TABLE users (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100),  
  email VARCHAR(100),  
  password VARCHAR(100),  
  role VARCHAR(50)  
);
```

Create TABLE bugs:

```
CREATE TABLE bugs (  
  bug_id INT AUTO_INCREMENT PRIMARY KEY,  
  bug_name VARCHAR(255) NOT NULL,  
  bug_type VARCHAR(100),  
  bug_priority VARCHAR(50),  
  project_name VARCHAR(100),  
  bug_level VARCHAR(50),  
  start_date DATE,  
  reported_by VARCHAR(100),  
  assigned_to VARCHAR(100),  
  severity VARCHAR(50),  
  status VARCHAR(50) DEFAULT 'Open',  
  date_reported DATE DEFAULT (CURRENT_DATE),  
  date_fixed DATE DEFAULT NULL  
);
```

Table: users

```
INSERT INTO users (name, email, password, role)  
VALUES  
( 'Admin 1', 'admin1@example.com', 'admin123', 'Admin'),  
( 'Admin 2', 'admin2@example.com', 'admin123', 'Admin'),  
( 'Admin 3', 'admin3@example.com', 'admin123', 'Admin'),  
( 'Admin 4', 'admin4@example.com', 'admin123', 'Admin'),  
( 'Admin 5', 'admin5@example.com', 'admin123', 'Admin'),  
( 'Admin 6', 'admin6@example.com', 'admin123', 'Admin'),  
( 'Admin 7', 'admin7@example.com', 'admin123', 'Admin'),  
( 'Developer A', 'devA@example.com', 'dev123', 'Developer'),  
( 'Developer B', 'devB@example.com', 'dev123', 'Developer'),  
( 'Developer C', 'devC@example.com', 'dev123', 'Developer'),  
( 'Developer D', 'devD@example.com', 'dev123', 'Developer'),  
( 'Developer E', 'devE@example.com', 'dev123', 'Developer'),  
( 'Developer F', 'devF@example.com', 'dev123', 'Developer'),  
( 'Developer G', 'devG@example.com', 'dev123', 'Developer'),  
( 'Tester 1', 'tester1@example.com', 'test123', 'Tester'),  
( 'Tester 2', 'tester2@example.com', 'test123', 'Tester'),  
( 'Tester 3', 'tester3@example.com', 'test123', 'Tester'),  
( 'Tester 4', 'tester4@example.com', 'test123', 'Tester'),  
( 'Tester 5', 'tester5@example.com', 'test123', 'Tester'),  
( 'Tester 6', 'tester6@example.com', 'test123', 'Tester'),  
( 'Tester 7', 'tester7@example.com', 'test123', 'Tester');  
INSERT INTO users (name, email, password, role) VALUES  
( 'Project Manager 1', 'pm1@example.com', 'pm123', 'Project Manager'),  
( 'Project Manager 2', 'pm2@example.com', 'pm123', 'Project Manager'),  
( 'Project Manager 3', 'pm3@example.com', 'pm123', 'Project Manager'),  
( 'Project Manager 4', 'pm4@example.com', 'pm123', 'Project Manager'),  
( 'Project Manager 5', 'pm5@example.com', 'pm123', 'Project Manager'),  
( 'Project Manager 6', 'pm6@example.com', 'pm123', 'Project Manager'),  
( 'Project Manager 7', 'pm7@example.com', 'pm123', 'Project Manager');
```

	id	name	email	password	role
	Filter	Filter	Filter	Filter	Filter
1	1	Admin 1	admin1@example.com	admin123	Admin
2	2	Admin 2	admin2@example.com	admin123	Admin
3	3	Admin 3	admin3@example.com	admin123	Admin
4	4	Admin 4	admin4@example.com	admin123	Admin
5	5	Admin 5	admin5@example.com	admin123	Admin
6	6	Admin 6	admin6@example.com	admin123	Admin
7	7	Admin 7	admin7@example.com	admin123	Admin
8	8	Developer A	devA@example.com	dev123	Developer
9	9	Developer B	devB@example.com	dev123	Developer
10	10	Developer C	devC@example.com	dev123	Developer
11	11	Developer D	devD@example.com	dev123	Developer
12	12	Developer E	devE@example.com	dev123	Developer
13	13	Developer F	devF@example.com	dev123	Developer
14	14	Developer G	devG@example.com	dev123	Developer
15	15	Tester 1	tester1@example.com	test123	Tester
16	16	Tester 2	tester2@example.com	test123	Tester
17	17	Tester 3	tester3@example.com	test123	Tester
18	18	Tester 4	tester4@example.com	test123	Tester
19	19	Tester 5	tester5@example.com	test123	Tester
20	20	Tester 6	tester6@example.com	test123	Tester
21	21	Tester 7	tester7@example.com	test123	Tester
22	22	Project Manager	pm1@example.com	pm123	Project Manager
23	23	Project Manager	pm2@example.com	pm123	Project Manager
24	24	Project Manager	pm3@example.com	pm123	Project Manager
25	25	Project Manager	pm4@example.com	pm123	Project Manager
26	26	Project Manager	pm5@example.com	pm123	Project Manager
27	27	Project Manager	pm6@example.com	pm123	Project Manager
28	28	Project Manager	pm7@example.com	pm123	Project Manager

Table: bugs

```
INSERT INTO bugs
(bug_name, bug_type, bug_priority, project_name, bug_level, start_date, reported_by, assigned_to, severity, status)
VALUES
('Slow page load on dashboard', 'Performance', 'Low', 'ECommerceApp', 'Minor', CURDATE(), 'Tester A', 'Developer A', 'Low', 'Fixed'),
('Data not saving in DB', 'Backend', 'High', 'InventorySystem', 'Critical', CURDATE(), 'Tester B', 'Developer B', 'High', 'Open'),
('Email validation fails', 'Validation', 'Medium', 'RegistrationApp', 'Major', CURDATE(), 'Tester C', 'Developer C', 'Medium', 'Open'),
('Search results incorrect', 'Logic', 'High', 'LibrarySystem', 'Major', CURDATE(), 'Tester D', 'Developer D', 'High', 'Fixed'),
('Broken image on homepage', 'UI', 'Low', 'PortfolioSite', 'Trivial', CURDATE(), 'Tester E', 'Developer E', 'Low', 'Assigned'),
('Payment gateway timeout', 'Network', 'High', 'ShoppingCart', 'Critical', CURDATE(), 'Tester F', 'Developer F', 'High', 'Fixed'),
('Password reset email not sent', 'Backend', 'Medium', 'UserPortal', 'Major', CURDATE(), 'Tester G', 'Developer G', 'Medium', 'Open'),
('Dark mode theme mismatch', 'UI', 'Low', 'SocialApp', 'Minor', CURDATE(), 'Tester A', 'Developer A', 'Low', 'Closed'),
('API returns 500 error', 'Backend', 'High', 'AnalyticsDashboard', 'Critical', CURDATE(), 'Tester B', 'Developer B', 'High', 'Fixed'),
('File upload exceeds limit', 'Validation', 'Medium', 'DocumentManager', 'Major', CURDATE(), 'Tester C', 'Developer C', 'Medium', 'Assigned'),
('Notification not displaying', 'Frontend', 'Low', 'ChatSystem', 'Minor', CURDATE(), 'Tester D', 'Developer D', 'Low', 'Fixed'),
('Report export broken', 'Backend', 'High', 'HRMS', 'Critical', CURDATE(), 'Tester E', 'Developer E', 'High', 'Open'),
('Session timeout too early', 'Logic', 'Medium', 'BankingApp', 'Major', CURDATE(), 'Tester F', 'Developer F', 'Medium', 'Closed'),
('Missing validation on signup', 'Validation', 'High', 'RegistrationApp', 'Major', CURDATE(), 'Tester G', 'Developer G', 'High', 'Assigned'),
('Image upload not working', 'Frontend', 'High', 'MediaPortal', 'Critical', CURDATE(), 'Tester A', 'Developer C', 'High', 'Fixed');
```

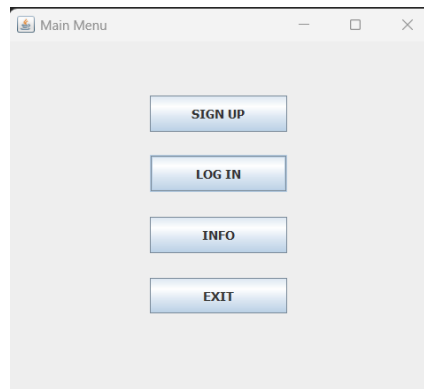
bug_id	bug_name	bug_type	bug_priority	project_name	bug_level	start_date	reported_by	assigned_to	severity	status	date_reported	date_fixed
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	Slow page load on dashboard	Performance	Low	ECommerceApp	Minor	2025-10-30	Tester A	Developer A	Low	Fixed	2025-10-30	(NULL)
2	Data not saving in DB	Backend	High	InventorySystem	Critical	2025-10-30	Tester B	Developer B	High	Open	2025-10-30	(NULL)
3	Email validation fails	Validation	Medium	RegistrationApp	Major	2025-10-30	Tester C	Developer C	Medium	Open	2025-10-30	(NULL)
4	Search results incorrect	Logic	High	LibrarySystem	Major	2025-10-30	Tester D	Developer D	High	Fixed	2025-10-30	(NULL)
5	Broken image on homepage	UI	Low	PortfolioSite	Trivial	2025-10-30	Tester E	Developer E	Low	Assigned	2025-10-30	(NULL)
6	Payment gateway timeout	Network	High	ShoppingCart	Critical	2025-10-30	Tester F	Developer F	High	Fixed	2025-10-30	(NULL)
7	Password reset email not sent	Backend	Medium	UserPortal	Major	2025-10-30	Tester G	Developer G	Medium	Open	2025-10-30	(NULL)
8	Dark mode theme mismatch	UI	Low	SocialApp	Minor	2025-10-30	Tester A	Developer A	Low	Closed	2025-10-30	(NULL)
9	API returns 500 error	Backend	High	AnalyticsDashboard	Critical	2025-10-30	Tester B	Developer B	High	Fixed	2025-10-30	(NULL)
10	File upload exceeds limit	Validation	Medium	DocumentManager	Major	2025-10-30	Tester C	Developer C	Medium	Assigned	2025-10-30	(NULL)
11	Notification not displaying	Frontend	Low	ChatSystem	Minor	2025-10-30	Tester D	Developer D	Low	Fixed	2025-10-30	(NULL)
12	Report export broken	Backend	High	HRMS	Critical	2025-10-30	Tester E	Developer E	High	Open	2025-10-30	(NULL)
13	Session timeout too early	Logic	Medium	BankingApp	Major	2025-10-30	Tester F	Developer F	Medium	Closed	2025-10-30	(NULL)
14	Missing validation on signup	Validation	High	RegistrationApp	Major	2025-10-30	Tester G	Developer G	High	Assigned	2025-10-30	(NULL)
15	Image upload not working	Frontend	High	MediaPortal	Critical	2025-10-30	Tester A	Developer C	High	Fixed	2025-10-30	(NULL)

OUTPUT SNAPS:

MAIN MENU

The Main Menu Window serves as the central navigation hub of the Bug Tracking System.

It provides users with an interface to SIGN UP , LOGIN, INFO and EXIT.

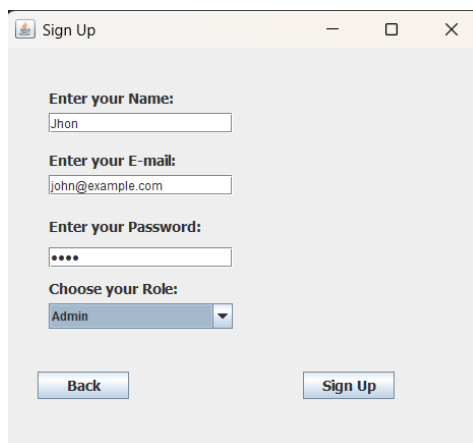
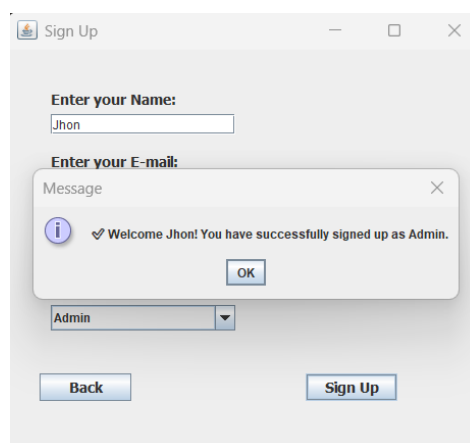


SIGN UP

The Sign-Up Window (also called Registration Window) allows new users to create an account in the Bug Tracking System.

SQL Query:

```
INSERT INTO users (username, email, password, role)
VALUES (?, ?, ?, ?);
```

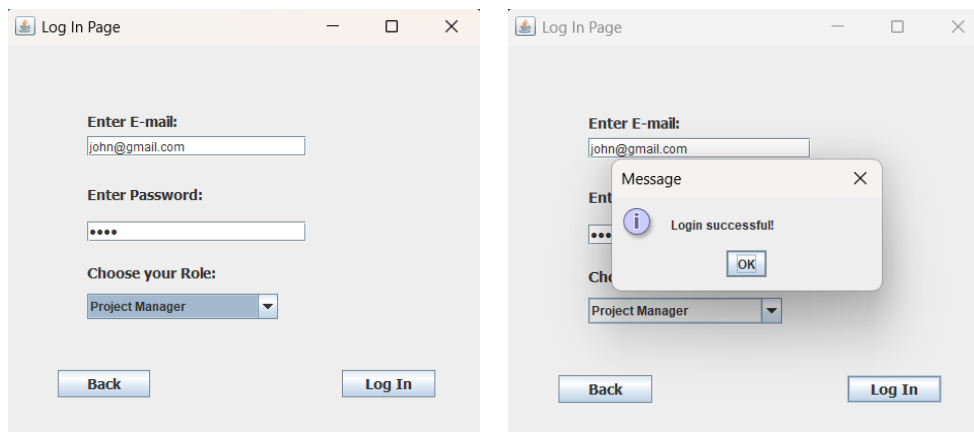
A screenshot of a window titled "Sign Up". The window has a light gray background and contains four input fields with labels: "Enter your Name:" (text box with "Jhon"), "Enter your E-mail:" (text box with "john@example.com"), "Enter your Password:" (password box with four dots), and "Choose your Role:" (dropdown menu with "Admin" selected). At the bottom, there are two blue buttons: "Back" and "Sign Up".A screenshot of a window titled "Sign Up" showing a success message. A modal dialog box is open in the center, titled "Message", with an information icon and the text "Welcome Jhon! You have successfully signed up as Admin." and an "OK" button. The background window is slightly dimmed, showing the same "Sign Up" form as the previous screenshot.

LOGIN

The Login Window is the entry point of the system. It authenticates users based on their username, password, and role, and redirects them to their respective dashboards.

SQL Query:

```
SELECT * FROM users WHERE username=? AND password=? AND role=?;
```

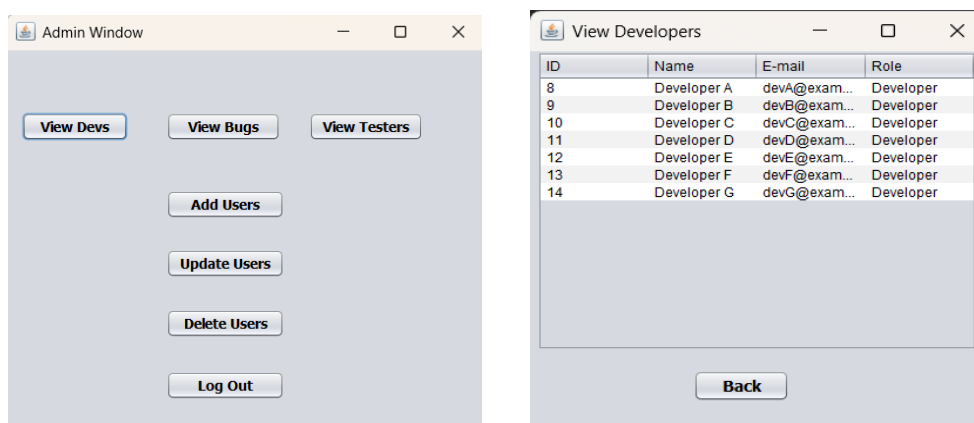


ADMIN

The Admin Window provides control over user management and system configuration. Admins can view registered users, add new accounts, and access developer/tester lists.

SQL Query:

```
SELECT * FROM users WHERE role='Developer';  
SELECT * FROM users WHERE role='Tester';  
INSERT INTO users (username, email, password, role) VALUES (?, ?, ?, ?);
```



Bug ID	Bug Name	Bug Type	Bug Priority	Project Name	Bug Level	Start Date	Reported By	Assigned To	Severity	Status
1	Slow page ...	Performance	Low	ECommerc...	Minor	2025-10-30	Tester A	Developer C	Low	Resolved
2	Data not sa...	Backend	High	InventorySy...	Critical	2025-10-30	Tester B	Developer B	High	Open
3	Email valid...	Validation	Medium	Registratio...	Major	2025-10-30	Tester C	Developer D	Medium	Resolved
4	Search res...	Logic	High	LibrarySyst...	Major	2025-10-30	Tester D	Developer D	High	Fixed
5	Broken ima...	UI	Low	PortfolioSite	Trivial	2025-10-30	Tester E	Developer F	Low	Open
6	Payment g...	Network	High	ShoppingC...	Critical	2025-10-30	Tester F	Developer F	High	Fixed
7	Password r...	Backend	Medium	UserPortal	Major	2025-10-30	Tester G	Developer G	Medium	Fixed
8	Dark mode...	UI	Low	SocialApp	Minor	2025-10-30	Tester A	Developer A	Low	Closed
9	API returns ...	Backend	High	AnalyticsDa...	Critical	2025-10-30	Tester B	Developer B	High	Fixed
10	File upload...	Validation	Medium	Document...	Major	2025-10-30	Tester C	Developer C	Medium	Fixed
11	Notification...	Frontend	Low	ChatSystem	Minor	2025-10-30	Tester D	Developer D	Low	Fixed
12	Report exp...	Backend	High	HRMS	Critical	2025-10-30	Tester E	Developer E	High	Open
13	Session tim...	Logic	Medium	BankingApp	Major	2025-10-30	Tester F	Developer F	Medium	Closed
14	Missing val...	Validation	High	Registratio...	Major	2025-10-30	Tester G	Developer G	High	Assigned
15	Image uplo...	Frontend	High	MediaPortal	Critical	2025-10-30	Tester A	Developer C	High	Fixed
16	Backgroun...	UI/UX	Medium	Online Sho...	Minor	2023-10-10	Tester3	DeveloperG	Medium	Fixed
17	Back butto...	Backend	Low	Bug Trackin...	Major	2022-10-10	Tester C	Developer A	Medium	In Progress
18	Incorrect S...	Performance	Medium	Online Quiz...	Minor	2021-09-04	Developer	Project Ma...	Medium	Open

ID	Name	E-mail
15	Tester 1	tester1@example...
16	Tester 2	tester2@example....
17	Tester 3	tester3@example....
18	Tester 4	tester4@example....
19	Tester 5	tester5@example....
20	Tester 6	tester6@example....
21	Tester 7	tester7@example....
33	Kushi	kushi@gmail.com

i) ADD USER

Add User

Enter your Name:

John

Enter your ID:

101

Enter your Password:

Enter your E-mail:

john@gmail.com

Choose your Role:

Project Manager

Back

Add

Add User

Enter your Name:

John

Enter your ID:

101

Enter your Password:

Enter your E-mail:

john@gmail.com

Choose your Role:

Project Manager

Back

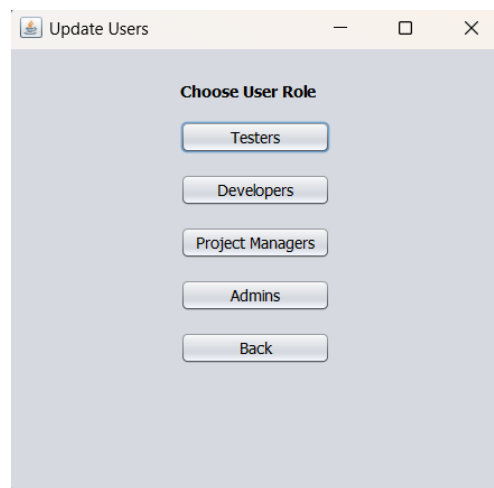
Add

Message

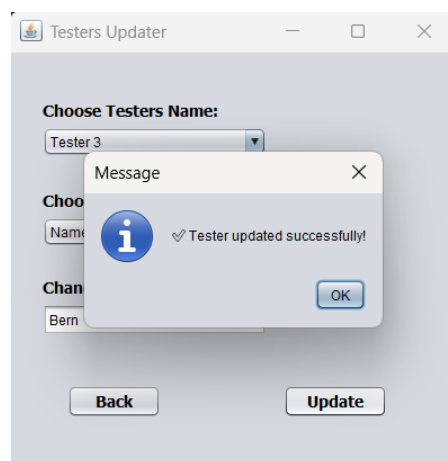
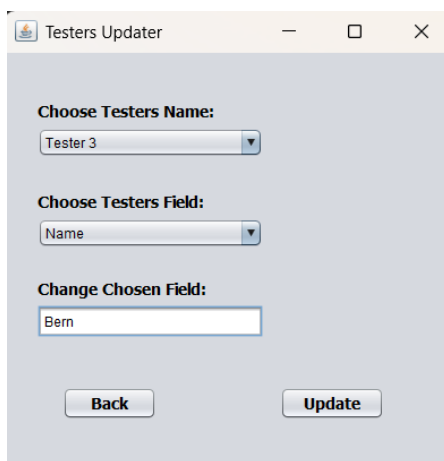
User Added Successfully!

OK

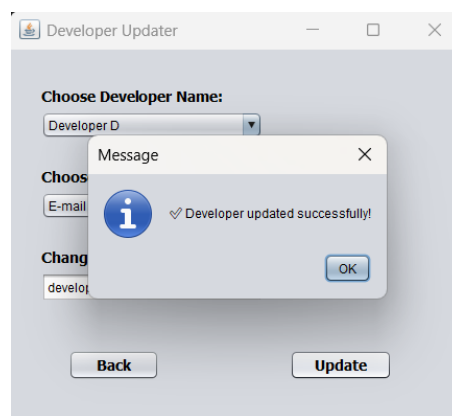
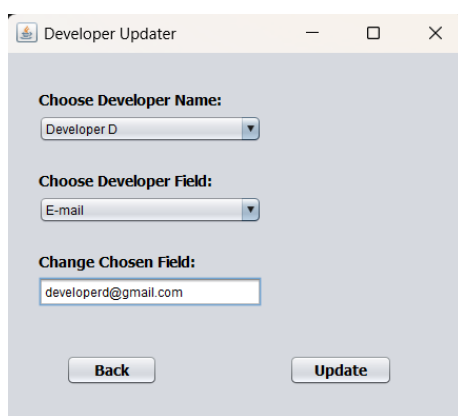
ii) UPDATE USER



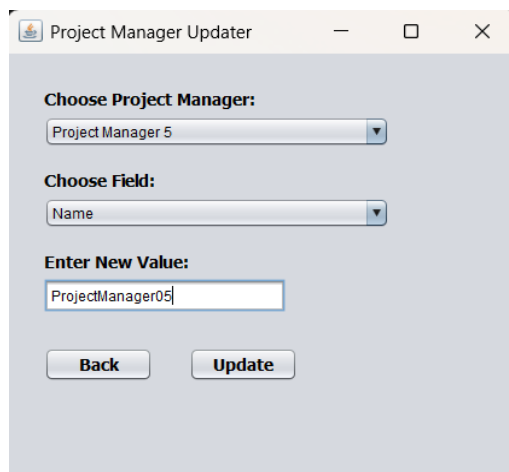
Tester Updater



Developer Updater



Project Manager Updater



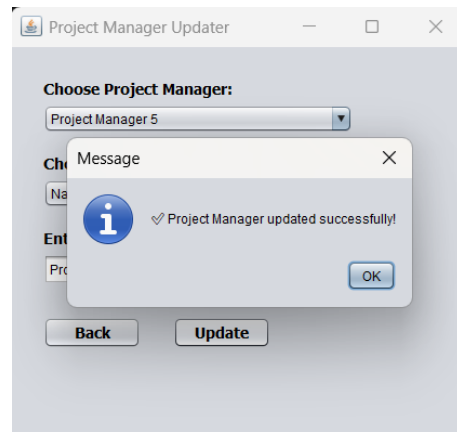
Project Manager Updater

Choose Project Manager:
Project Manager 5

Choose Field:
Name

Enter New Value:
ProjectManager05

Back Update



Project Manager Updater

Choose Project Manager:
Project Manager 5

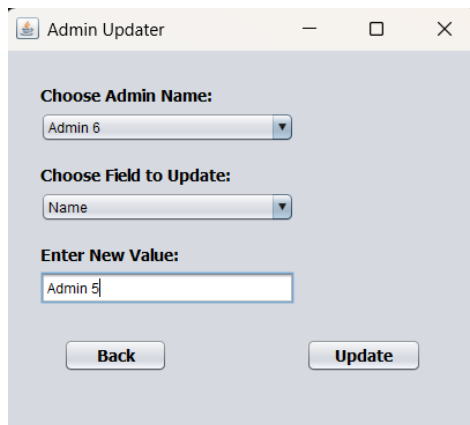
Choose Field:
Name

Enter New Value:
ProjectManager05

Back Update

Message
✔ Project Manager updated successfully!
OK

Admin Updater



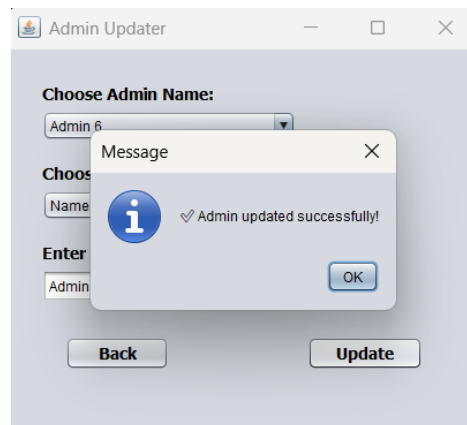
Admin Updater

Choose Admin Name:
Admin 6

Choose Field to Update:
Name

Enter New Value:
Admin 5

Back Update



Admin Updater

Choose Admin Name:
Admin 6

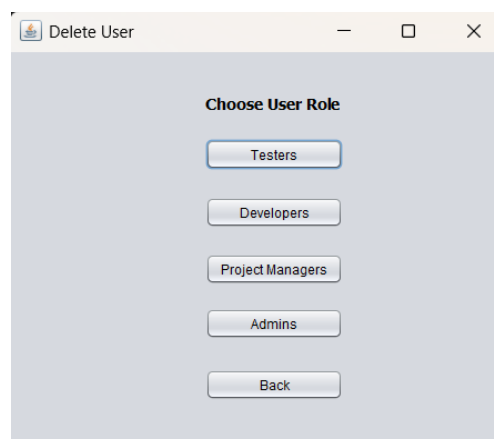
Choose Field to Update:
Name

Enter New Value:
Admin 5

Back Update

Message
✔ Admin updated successfully!
OK

iii) DELETE USER



Delete User

Choose User Role

Testers

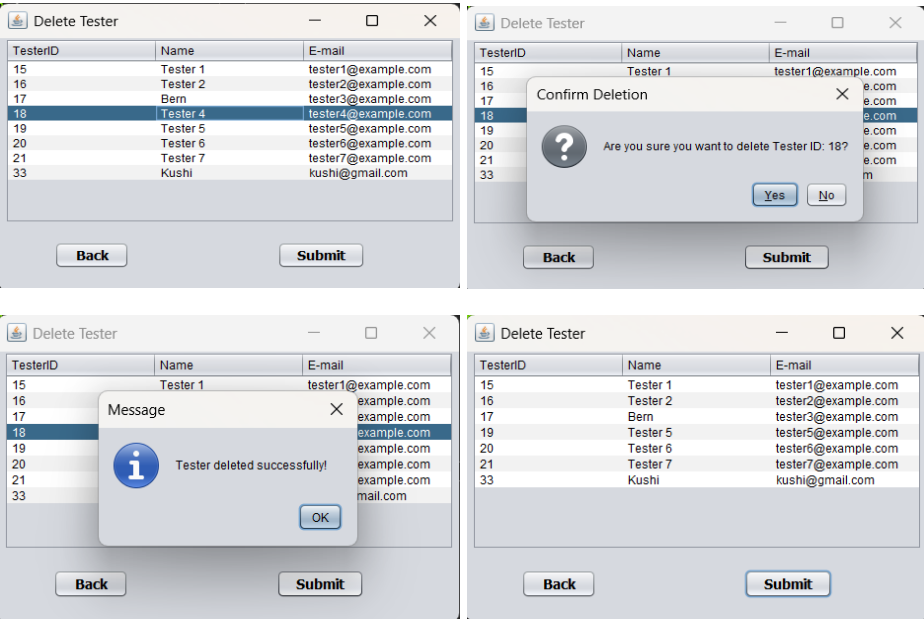
Developers

Project Managers

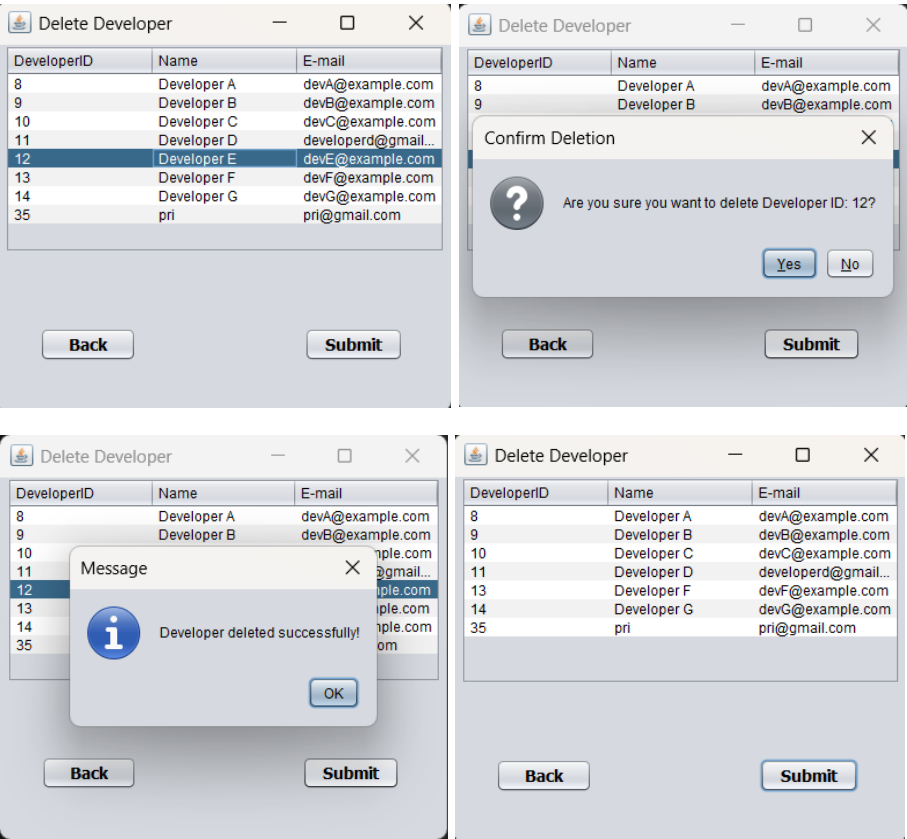
Admins

Back

Delete Tester



Delete Developer



Delete Project Manager

Delete Project Manager

PM ID	Name	E-mail
22	Project Manager 1	pm1@example.com
23	Project Manager 2	pm2@example.com
25	Project Manager 4	pm4@example.com
26	ProjectManager05	pm5@example.com
27	Project Manager 6	pm6@example.com
28	Project Manager 7	pm7@example.com
36	101	john@gmail.com

Back

Submit

Delete Project Manager

PM ID	Name	E-mail
22	Project Manager 1	pm1@example.com
23	Project Manager 2	pm2@example.com
25	Project Manager 4	pm4@example.com
26	ProjectManager05	pm5@example.com
27	Project Manager 6	pm6@example.com
28	Project Manager 7	pm7@example.com
36	101	john@gmail.com

Confirm Deletion

?

Are you sure you want to delete Project Manager ID: 27?

Yes

No

Back

Submit

Delete Project Manager

PM ID	Name	E-mail
22	Project Manager 1	pm1@example.com
23	Project Manager 2	pm2@example.com
25	Project Manager 4	pm4@example.com
26	ProjectManager05	pm5@example.com
27	Project Manager 6	pm6@example.com
28	Project Manager 7	pm7@example.com
36	101	john@gmail.com

Message

i

Project Manager deleted successfully!

OK

Back

Submit

Delete Project Manager

PM ID	Name	E-mail
22	Project Manager 1	pm1@example.com
23	Project Manager 2	pm2@example.com
25	Project Manager 4	pm4@example.com
26	ProjectManager05	pm5@example.com
28	Project Manager 7	pm7@example.com
36	101	john@gmail.com

Back

Submit

Delete Admin

Delete Admin

AdminID	Name	E-mail
1	Admin 1	admin1@example.c...
2	Admin 2	admin2@example.c...
3	Admin 3	admin3@example.c...
4	Admin 4	admin4@example.c...
6	Admin 5	admin6@example.c...
7	Admin 7	admin7@example.c...
29	rakshu	rakshu@gmail.com
30	alice	alice@gmail.com
31	Biradar	biradar@gmail.com
32	Shreya	shreya@gmail.com
34	Jhon	john@example.com

Back

Submit

Delete Admin

AdminID	Name	E-mail
1	Admin 1	admin1@example.c...
2	Admin 2	admin2@example.c...
3	Admin 3	admin3@example.c...
4	Admin 4	admin4@example.c...
6		
7		
29		
30		
31		
32		
34		

Confirm Deletion

?

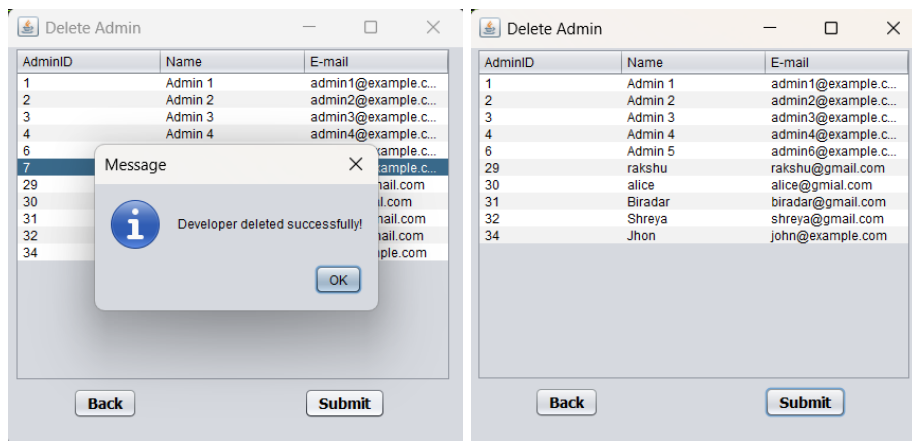
Are you sure you want to delete Developer ID: 7?

Yes

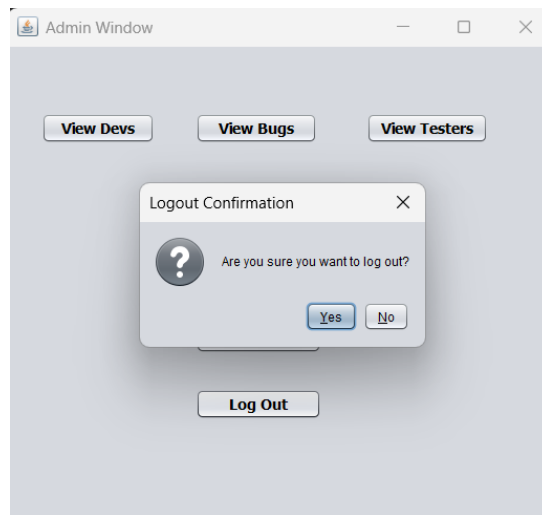
No

Back

Submit

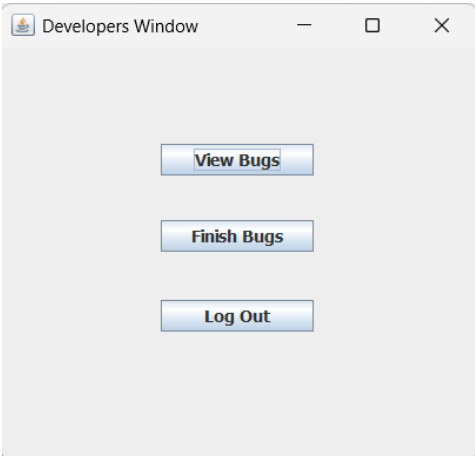


iv) LOGOUT



DEVELOPERS WINDOW

The Developer Dashboard allows developers to view bugs assigned to them and update their status after resolution.



View All Bugs

View All Bugs - Developer										
Bug ID	Bug Name	Bug Type	Bug Priority	Project Name	Bug Level	Start Date	Reported By	Assigned To	Severity	Status
1	Slow page l...	Performance	Low	ECommerce...	Minor	2025-10-30	Tester A	Developer C	Low	Resolved
2	Data not sav...	Backend	High	InventorySys...	Critical	2025-10-30	Tester B	Developer B	High	Open
3	Email validat...	Validation	Medium	Registration...	Major	2025-10-30	Tester C	Developer D	Medium	Resolved
4	Search resul...	Logic	High	LibrarySystem	Major	2025-10-30	Tester D	Developer D	High	Fixed
5	Broken imag...	UI	Low	PortfolioSite	Trivial	2025-10-30	Tester E	Developer F	Low	Open
6	Payment gat...	Network	High	ShoppingCart	Critical	2025-10-30	Tester F	Developer F	High	Fixed
7	Password re...	Backend	Medium	UserPortal	Major	2025-10-30	Tester G	Developer G	Medium	Fixed
8	Dark mode t...	UI	Low	SocialApp	Minor	2025-10-30	Tester A	Developer A	Low	Closed
9	API returns 5...	Backend	High	AnalyticsDas...	Critical	2025-10-30	Tester B	Developer B	High	Fixed
10	File upload ...	Validation	Medium	DocumentM...	Major	2025-10-30	Tester C	Developer C	Medium	Fixed
11	Notification ...	Frontend	Low	ChatSystem	Minor	2025-10-30	Tester D	Developer D	Low	Fixed
12	Report expo...	Backend	High	HRMS	Critical	2025-10-30	Tester E	Developer E	High	Open
13	Session time...	Logic	Medium	BankingApp	Major	2025-10-30	Tester F	Developer F	Medium	Closed
14	Missing vali...	Validation	High	Registration...	Major	2025-10-30	Tester G	Developer G	High	Assigned
15	Image uploa...	Frontend	High	MediaPortal	Critical	2025-10-30	Tester A	Developer C	High	Fixed
16	Background ...	UI/UX	Medium	Online Shop...	Minor	2023-10-10	Tester3	DeveloperG	Medium	Fixed
17	Back button ...	Backend	Low	Bug Trackin...	Major	2022-10-10	Tester C	Developer A	Medium	In Progress
18	Incorrect SQ...	Performance	Medium	Online Quiz ...	Minor	2021-09-04	Developer	Project Man...	Medium	Open

Back

Finish Bug

Finish Bug

Bug ID:

12

Bug Name:

Report export broken

Assigned To:

Developer E

New Status:

Fixed

Date Fixed:

2025-10-30

Back

Submit

Finish Bug

Bug ID:

12

Bug Name:

Report export broken

Assigned To:

Developer E

New Status:

Fixed

Date Fixed:

2025-10-30

Back

Submit

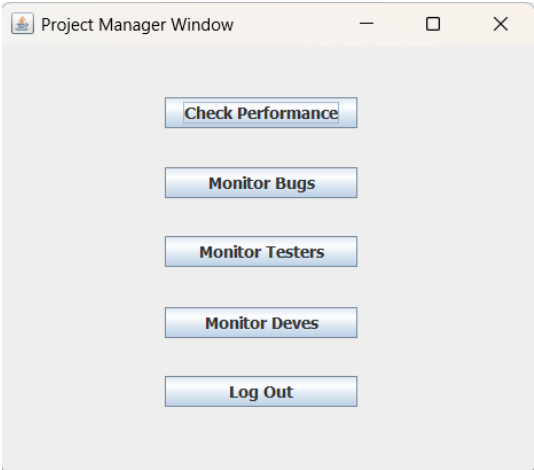
Message

Bug updated successfully.

OK

PROJECT MANAGER WINDOW

The Project Manager Window serves as the main dashboard for users with the Project Manager role. It provides access to key management functionalities assigning bugs to developers, tracking progress, and viewing bug statistics.



Check Performance



Monitor Bugs

Monitor Bugs								
Bug ID	Title	Type	Severity	Priority	Project	Assigned To	Status	
1	Slow page load...	Performance	Minor	Low	ECommerceApp	Developer A	Fixed	
2	Data not saving...	Backend	Critical	High	InventorySystem	Developer B	Open	
3	Email validatio...	Validation	Major	Medium	RegistrationApp	Developer C	Open	
4	Search results ...	Logic	Major	High	LibrarySystem	Developer D	Fixed	
5	Broken image ...	UI	Trivial	Low	PortfolioSite	Developer E	Assigned	
6	Payment gatew...	Network	Critical	High	ShoppingCart	Developer F	Fixed	
7	Password rese...	Backend	Major	Medium	UserPortal	Developer G	Open	
8	Dark mode the...	UI	Minor	Low	SocialApp	Developer A	Closed	
9	API returns 500...	Backend	Critical	High	AnalyticsDashb...	Developer B	Fixed	
10	File upload exc...	Validation	Major	Medium	DocumentMan...	Developer C	Assigned	
11	Notification not ...	Frontend	Minor	Low	ChatSystem	Developer D	Fixed	
12	Report export b...	Backend	Critical	High	HRMS	Developer E	Open	
13	Session timeo...	Logic	Major	Medium	BankingApp	Developer F	Closed	
14	Missing validati...	Validation	Major	High	RegistrationApp	Developer G	Assigned	
15	Image upload ...	Frontend	Critical	High	MediaPortal	Developer C	Fixed	

Monitor Testers

Monitor Testers		
ID	Name	Email
15	Tester 1	tester1@example.com
16	Tester 2	tester2@example.com
17	Tester 3	tester3@example.com
18	Tester 4	tester4@example.com
19	Tester 5	tester5@example.com
20	Tester 6	tester6@example.com
21	Tester 7	tester7@example.com

Monitor Developers

Monitor Developers		
ID	Name	Email
8	Developer A	devA@example.com
9	Developer B	devB@example.com
10	Developer C	devC@example.com
11	Developer D	devD@example.com
12	Developer E	devE@example.com
13	Developer F	devF@example.com
14	Developer G	devG@example.com

Log Out

Project Manager Window

Check Performance

Log Out

Are you sure you want to log out?

OKCancel

Monitor Deves

Log Out

Project Manager Window

Check Performance

Message

You have been logged out successfully!

OK

Monitor Deves

Log Out

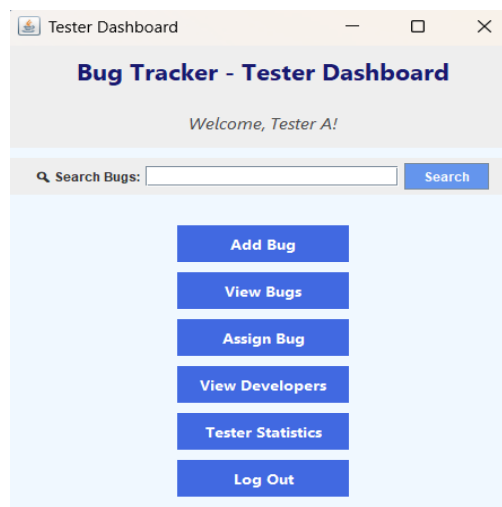
TESTER WINDOW

The Tester Dashboard allows testers to report new bugs, search bugs, and view all submitted issues.

SQL Query:

```
SELECT * FROM bugs WHERE bug_name LIKE ?;
```

```
SELECT * FROM bugs WHERE created_by=?;
INSERT INTO bugs (bug_name, description, severity, status, created_by, created_date)
VALUES (?, ?, ?, 'Open', ?, ?);
```



Add Bug

A screenshot of a web application window titled "Add New Bug Report". The form contains several input fields and dropdown menus: "Bug Name" (text input), "Bug Type" (text input), "Project Name" (text input), "Priority" (dropdown menu), "Bug Level" (dropdown menu), "Start Date (yyyy-MM-dd)" (text input), "Reported By" (text input), "Assigned To" (text input), "Severity" (dropdown menu), and "Status" (dropdown menu). The form has a "Back" button, an "Add Bug" button, and a "Clear" button at the bottom. A modal message box is displayed over the form, showing a green checkmark and the text "Bug added successfully!".

View All Bugs

View All Bugs										
Bug ID	Bug Name	Bug Type	Bug Priority	Project Name	Bug Level	Start Date	Reported By	Assigned To	Severity	Status
1	Slow page l...	Performance	Low	ECommerce...	Minor	2025-10-30	Tester A	Developer A	Low	Fixed
2	Data not sav...	Backend	High	InventorySys...	Critical	2025-10-30	Tester B	Developer B	High	Open
3	Email validat...	Validation	Medium	Registration...	Major	2025-10-30	Tester C	Developer C	Medium	Open
4	Search resul...	Logic	High	LibrarySystem	Major	2025-10-30	Tester D	Developer D	High	Fixed
5	Broken imag...	UI	Low	PortfolioSite	Trivial	2025-10-30	Tester E	Developer E	Low	Assigned
6	Payment gat...	Network	High	ShoppingCart	Critical	2025-10-30	Tester F	Developer F	High	Fixed
7	Password re...	Backend	Medium	UserPortal	Major	2025-10-30	Tester G	Developer G	Medium	Open
8	Dark mode t...	UI	Low	SocialApp	Minor	2025-10-30	Tester A	Developer A	Low	Closed
9	API returns 5...	Backend	High	AnalyticsDas...	Critical	2025-10-30	Tester B	Developer B	High	Fixed
10	File upload ...	Validation	Medium	DocumentM...	Major	2025-10-30	Tester C	Developer C	Medium	Assigned
11	Notification ...	Frontend	Low	ChatSystem	Minor	2025-10-30	Tester D	Developer D	Low	Fixed
12	Report expo...	Backend	High	HRMS	Critical	2025-10-30	Tester E	Developer E	High	Open
13	Session time...	Logic	Medium	BankingApp	Major	2025-10-30	Tester F	Developer F	Medium	Closed
14	Missing vali...	Validation	High	Registration...	Major	2025-10-30	Tester G	Developer G	High	Assigned
15	Image uploa...	Frontend	High	MediaPortal	Critical	2025-10-30	Tester A	Developer C	High	Fixed
16	Logout Redi...	Functional B...	Medium	Bug Pro Ma...	Major	2021-11-29	Tester D	Developer F	Medium	Open

Back

Assign Bug

Assign Bug

Select Bug:

5 - Broken image on homena...

Assign To:

Developer D

Status:

Resolved

Assign Bug

Cancel

Back

Assign Bug

Message

✔ Bug assigned successfully!

OK

Assign Bug

Cancel

Back

View Developers

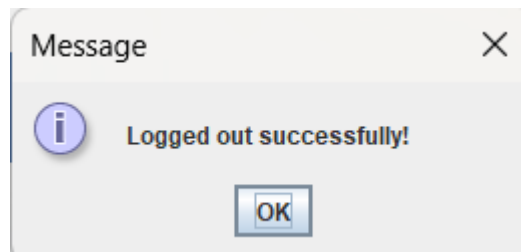
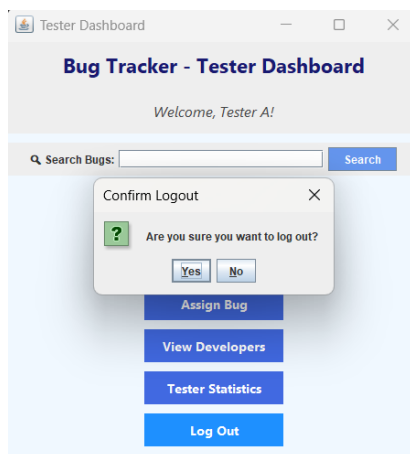
View Developers			
ID	Name	E-mail	Role
8	Developer A	devA@exampl...	Developer
9	Developer B	devB@exampl...	Developer
10	Developer C	devC@exampl...	Developer
11	Developer D	devD@exampl...	Developer
12	Developer E	devE@exampl...	Developer
13	Developer F	devF@exampl...	Developer
14	Developer G	devG@exampl...	Developer

Back

Tester Statistics



Log Out



FUTURE ENHANCEMENTS

- Implement email notifications when bugs are assigned or resolved.
- Add analytics and reports (e.g., bugs by severity or developer).
- Include password encryption for secure authentication.
- Convert into a web-based system using JavaFX or Spring Boot.
- Provide multi-project support.

CONCLUSION

The Bug Tracking System provides a comprehensive platform for managing the bug lifecycle. By leveraging Java Swing for the interface and MySQL for persistent storage, it ensures data integrity, user interactivity, and efficient workflow. This system demonstrates strong concepts in Object-Oriented Programming, Database Connectivity (JDBC), and GUI Design, making it a practical and educational project for understanding real-world software maintenance systems.
